

Project — See the Shape of a Neural Code

Time to finish: 1 day (about 4-6 hours, working alone or in a pair).

What you will do

You are given the responses of many neurons to a set of visual stimuli. Your job is to answer two questions:

1. How many "directions" does the neural code really use?
2. Can we read the stimulus back from the neural activity?

Background

- PCA (Principal Component Analysis) finds the main axes of variation in the data. If most of the variance sits in a few axes, the code is low-dimensional.
- Linear decoding means training a simple regression to predict the stimulus from the neural activity.

Data

In `shared/data/`:

- `population_responses.npz` — one matrix `X` of shape `(n_trials, n_neurons)`.
- `stimulus_grid.csv` — the stimulus (a 2-D value) shown on each trial.

Steps

3. Load and look. Open `student/notebooks/00_starter.ipynb`. Load the response matrix `X` and the stimulus grid `Y`. Print their shapes.
4. Run PCA. Use `sklearn.decomposition.PCA` on `X`. Plot the variance explained by each component (a bar plot). How many components explain 80% of the variance? That is your linear dimensionality estimate.
5. Project and plot. Take the first 2 principal components and make a scatter plot, colouring each point by the stimulus value. Does the picture look organised (nice colour gradient) or random?
6. Train a linear decoder. Use `sklearn.linear_model.Ridge` to predict `Y` from `X`. Split the data 80/20 (train/test) with `train_test_split`. Report the R^2 on the test set.

What to hand in

- Your filled notebook.
- The variance-explained bar plot.
- The 2-D PCA scatter plot coloured by stimulus.
- The R^2 of your decoder on the test set.
- A short paragraph (3-4 sentences): is the code low-dimensional? Does the decoder work well?

If you finish early: try predicting each stimulus dimension separately and see which one is easier to decode.